Pointers in C

Direct Memory Manipulation & Logic

Definitions • Dereferencing • Pointer Arithmetic

1. What is a Pointer?

A **pointer** is a variable that stores the memory address of another variable instead of storing a value directly.

Syntax:

```
data_type *pointer_name;
```

```
int a = 10;
int *p = &a; // p stores address of a
```

Dereferencing: Using the * operator to access the value at the address stored in the pointer.

2. Essential Operators

Operator	Name	Description
`&`	Address-of	Returns the memory address of a variable.
`*`	Dereference	Used to access the value at the address held by a pointer.

```
printf("%d", a);  // Output: 10
printf("%d", *p); // Output: 10 (Accesses value at address)
```

3. Pointer Arithmetic

Arithmetic operations like + and - behave differently with pointers. When you add 1 to an `int` pointer, it moves forward by the size of an integer (usually 4 bytes).

```
int *ptr = &a;
ptr++; // Moves to next memory block based on data type
```

Types of Pointers:

- **Null Pointer:** `int *p = NULL;` (Points to nothing).
- **Void Pointer:** `void *p;` (Generic pointer type).
- **Wild Pointer:** Uninitialized pointer (dangerous).

4. Pointers with Arrays

An array name acts as a constant pointer to its first element. This is why pointers are extremely efficient for traversing arrays.

```
int arr[3] = {10, 20, 30};
int *ptr = arr; // Same as &arr[0]

for(int i=0; i<3; i++) {
    printf("%d ", *(ptr + i)); // Accessing using pointer
}
```

5. Common Pitfalls

- **Uninitialized Pointers:** Using a pointer before assigning an address (leads to Segmentation Fault).
- **Dangling Pointers:** A pointer pointing to a memory location that has been deleted or freed.
- **Size Mismatch:** Assigning a `float` address to an `int*` without proper casting.

Pro Tip: Always initialize pointers to `NULL` if you don't have an address to assign immediately.

Module Complete

Pointers are the "Power Tool" of C. Mastering them allows for dynamic memory management and hardware-level programming.

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